# RISHI BABU

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## EDUCATION

**Michigan Technological University** Houghton, MI
*Ph.D. Physics* Feb 2024
Dissertation: Study of Particle Accelerators in the Universe with the HAWC Observatory
Dr. Petra Heuntemeyer

**Michigan Technological University** Houghton, MI
*M.S, Physics* April 2021

**National Institute of Technology, Calicut** Calicut, India
Bachelor of Technology, Engineering Physics *May 2017*

## TECHNICAL SKILLS

**Programming Languages:** Python, SQL, R, C++, MATLAB, Java
**Frameworks & Tools:** PyTorch, TensorFlow, JAX, CUDA, TensorRT, JupyterLab, Linux/Unix, Git/GitHub
**Scientific Computing:** HPC & Parallel Computing, Distributed Computing, SLURM
**Machine Learning: RNN<** Graph Neural Networks, Transformer Models, Deep Learning, GPU Accelerated Computing

## RESEARCH EXPERIENCE

**Michigan Technological University** Houghton, MI
*Graduate Researcher* Sept 2019 – Feb 2024

- Led high impact analytics projects with ambiguous objectives using R programming, such as separating signals from noisy backgrounds, transforming them into clearly scoped workflows with measurable metrics.
- Designed and deployed a Python+Unix based automation pipeline on HPC clusters, to replace manual data processing workflows, reducing human intervention by 60\% and significantly increasing efficiency and reproducibility.
- Built a detector electronics calibration tool with C++ that runs once a month calibrating the data acquisition system for the HAWC experiment, ensuring accurate measurement of signals.
- Applied Bayesian and MLE methods to detect hidden phenomenology in astrophysical data - drawing parallel to extracting latent creative insights using complex satellite data from NASA.

## PROFESSIONAL EXPERIENCE

**MICHIGAN STATE UNIVERSITY** Lansing, MI
*Postdoctoral Research Associate* Feb 2024 – Present

- Currently leading the planning and execution of HAWC Collaboration data releases, including coordination of analysis readiness, quality control, metadata standards, and documentation.
- Developer and maintainer of Python/Jupyter notebook workflows for reproducible analysis for the collaboration and public data release
- Managed distributed workflows for HPC systems for terabytes simulation requests and data analysis tasks for the collaboration.
- Built, trained and optimized physics based AI models with JAX and CUDA for GPU-accelerated data reconstruction on HPC clusters, cutting runtime from 15 minutes to milliseconds per epoch.
- Developed modular Python and SQL workflows for integrating and analyzing astronomical datasets for multiple experiments and check for reproducibility.
- Developer for the astromodels package, contributing to code optimization, maintenance, and scientific analysis workflows.
- Developed statistical models for multi-wavelength analysis of unidentified objects.
- Built image-processing based object detection and classification using clustering and watershed segmentation algorithms for astronomical maps, under support by Open Science Fellowship by NASA and Michigan State University.
- Led workshops and mentored analysts on statistical techniques, model designs and effectively communicated complex insights to a broader audience in the field.

## RESEARCH EXPERIENCE

**IceCube Collaboration Member ( Neutrino Astrophysics )**
*Michigan State University* *Feb* 2024 – Present

- Multi-Messenger Search for neutrinos from unidentified PeV gamma-ray with HAWC & IceCube Observatory
- Search for neutrinos  from star-forming galaxies based on the gamma-ray observations by Fermi-LAT.
- High energy neutrino track-like event reconstruction with physics based neural networks
- Search for tau neutrino interaction and decay in 12 years of IceCube Data using Machine Learning

**HAWC Collaboration Member ( Gamma-ray Astrophysics )**
*Michigan State University & Michigan Technological University* 2020 – Present

- Developer on astromodels python package
- Developer of diffuse gamma-ray emission models used for analysis by the HAWC Collaboration
- Developer of fast image-processing based source detection pipeline for realtime monitoring and archival data analysis pipeline
- Multi-wavelength and Multi-messenger analysis of ultra-high energy gamma-ray emission
- High energy neutrino track-like event reconstruction with physics based neural networks

## TEACHING EXPERIENCE

**Michigan State University** Lansing, MI
*Part-time Lecturer* *May* 2025 – Present

- Substitute teacher for PH294 (Honors) - Electricity and Magnetism

**Michigan Technological University** Houghton, MI
*Graduate Teaching Assistant* 2020 – 2024

- Class Instructor for PH1600 undergraduate Astronomy lab for 3 years
- Lab instructor for Honors Class PH1361 and PH2261, Intro Experimental Physics
- Teaching Assistant and substitute teacher for graduate Astrophysics class, PH4620 and PH4630

## LEADERSHIP AND SERVICE

- Public Data Release Lead, HAWC Collaboration 2026
- Data & Algorithms Working Group Leader, HAWC Collaboration 2025-Present
- Galactic Science Working Group Leader, HAWC Collaboration 2025
- Conference Session Chair, American Physical Society 2023 – 2025
- Department Representative, Graduate Student Government, Michigan Technological University 2020 – 2021
- Department Representative, Students Affairs Council, National Institute of Technology 2014 – 2017

## GRANTS EXPERIENCE

- **NSF Windows of the Universe: The era of Multi-messenger Astrophysics (WoU-MMA)} ,** Michigan State University (2024-2026)
- **NASA Fermi Guest Investigator Program**, Cycle 18, Co-PI (Accepted) 2025

## AWARDS & HONORS

- *Open Scholarship Fellowship, Michigan State University/NASA* *2025*
- *Dean's Award for Outstanding Scholarship, Michigan Technological University* *2024*
- *Outstanding Graduate Teaching Award, Michigan Technological University* *2024*
- *Henes Graduate Research Fellow, Michigan Technological University* *2024*
- *American Physical Society travel grant, APS* *2022 – 2023*
- *Recognition for Exceptional Teaching by the Provost, Michigan Technological University* *2021*

## PRESS COVERAGE

- *Press coverage for publication:* Observation of the Galactic Center PeVatron beyond 100 TeV with HAWC. *2025*
    - *Featured in* Los Alamos National Laboratory press release
    - *Featured in* CERN Courier, Universe Today *and* ScienceAlert

## PEER REVIEW

- Peer Review Panelist, **Fermi Gamma-Ray Space Telescope Guest Investigator Program**, NASA. 2025
- Peer Review Panelist, **NASA Citizen Science Seed Funding Program** (CSSRP), NASA 2025

## RESEARCH SUPERVISION AND MENTORSHIP

- Graduate Students
    - Alejandra Granados (MSU) 2025
    - Brandon Henke (MSU) 2025
    - Finn Mayhew (MSU) 2025
    - Joselyn Winda (MSU) 2025
    - Ian Herzog (MSU), now working at Niowave Inc 2025
    - Hitesh Malhotra (MSU) 2025
    - Manish Khanal ( University of Utah) 2025
- Undergraduate Students
    - Palmer Wentworth (MSU) 2025
    - Ved Vinod (MSU) 2025

## OUTREACH EFFORTS

- IceCube MasterClass for High School Students(MSU) 2025 - 2026
- Michigan State University Science Festival 2025 - 2026
- Astronomical Horizons Public Lecture Series, Abrams Planetarium 2025

## SEMINARS, COLLOQUIA AND SELECTED CONFERENCE TALKS

**Scientific and Technological Research Council of Türkiye, 2026,**
Invited Lecture "Multi-messenger Astrophysics with Neutrinos",Ankara, Turkey.

**Wright State University, 2025,**
Invited Lecture ”Understanding the Enigmatic Particle Accelerators in the Universe with the Multi-Messenger observations", Ohio, USA.

**Michigan State University, 2025,**
Invited Colloquium Talk "Understanding the Enigmatic Particle Accelerators in the Universe with the Multi-Messenger observations", Michigan, USA.

**Lepton-Photon Conference, 2025,**
"Multi-wavelength Analysis of Unassociated 1LHAASO Sources using HAWC Observatory", Madison, Wisconsin.

**ICRC, 2025 (Virtual),**
"Improving Gamma-ray Source Searches with Image Processing", Geneva, Switzerland.

**American Physical Society, March Meeting, 2025,**
"Correlation of Neutrino Emission with Ultra-High Energy Gamma-rays from LHAASO Dark Sources using IceCube and HAWC Observatories", California, USA.

**TeV Particle Astrophysics Conference, 2024**
“Analysis Results of HAWC Data Corresponding to 1LHAASO Sources Exclusively Linked to Fermi-LAT detected GeV Gamma Rays", Chicago, USA..

**11th International Fermi Symposium, 2024,**
"HAWC Study of the extended TeV emission from HESS J1813-178", Maryland, USA.

**ICRC, 2023 ,**
"High energy gamma ray emission from HESSJ1809-193: Morphological and Spectral studies with HAWC", Nagoya, Japan.

**American Physical Society, March Meeting, 2023,**
"HAWC Analysis of HESS J1809-193: a potential hadronic PeVatron?", Minneapolis, USA.

**American Physical Society, March Meeting, 2022,**
"HAWC Study of HESS J1809-193 - a PeVatron candidate in a source rich region", New York, USA.

## PUBLICATIONS

Primary Author

- A. Albert et al, "**Observation of the Galactic Center PeVatron Beyond 100 TeV with HAWC**", The Astrophysical Journal, vol. 973, no. 1, IOP, 2024.
- A. Albert et al,"**TeV Analysis of a Source Rich Region with HAWC Observatory: Is HESS J1809-193 a Potential Hadronic PeVatron?**" The Astrophysical Journal, Volume 972, 2024
- R. Babu, "**Improving Gamma-ray Source Searches with Fast Fourier Transforms and Image Processing**", PoS (ICRC2025) 557 (2025). DOI: 10.22323/1.501.0557
- A. Albert et al, "**HAWC Study on the Ultra-High-Energy Gamma-Ray Emissions from the Pulsar Wind Nebula G32.64+0.53**", Submitted to ApJ, 02 2026 (arXiv:2603.19721, 2026)

As a Contributing Author

- R. Alfaro et al,"**Ultra-high-energy gamma-ray bubble around microquasar V4641 Sgr**", Nature, vol. 634, no. 8034, pp. 557–560, 2024.
- R. Alfaro et al, "**Testing the molecular cloud paradigm for ultra-high-energy gamma ray emission from the direction of SNR G106.3+2.7**", Astronomy and Astrophysics, vol. 691, 2024.
- R. Alfaro et al, "**Analysis of the Emission and Morphology of the Pulsar Wind Nebula Candidate HAWC J2031+415**", The Astrophysical Journal, Volume 975, 2024.

In Preparation as Primary/Corresponding Author

- IceCube-HAWC, "**IceCube Search for Neutrino Emission from the Galactic Center detected by the HAWC Observatory**", The Astrophysical Journal, 2026 (Under review).
- HAWC, "**HAWC Detection of 1LHAASO J2027+3657 and its Connection to the Star-Forming Region S106**",
- The Astrophysical Journal, 2026 (Under review).
- HAWC, "**Identifying Ultra-High Energy Sources with Image Processing for Wide-Field Gamma-ray Survey Instruments and application to the HAWC Observatory**", ( Under Collaboration review ), To be submitted to ApJ, 03 2026.
- IceCube, "**Search for Neutrino Tracks from Starburst-Galaxies using 12 years of IceCube Data**", ( Under Collaboration review ), To be submitted to ApJ, 05 2026.
- IceCube-HAWC, "**Search for Neutrino Tracks and gamma-rays from LHAASO unidentified sources using IceCube and HAWC Observatories**", ( Under Collaboration review ), to ApJ, 06 2026.